Forces & Equilibrium

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Solve the inequality $x^2 + 17x + 72 > 0$.

The quadratic factorises as $(x + 9)(x + 8)$ with roots -9 and -8; since the parabola opens upward, the expression is > 0 outside the roots, giving $x < -9$ or $x > -8$.

2 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

A block rests on a rough horizontal surface with normal reaction $N = 42$ N. The coefficient of friction is $\mu = 7/20$. Find the limiting frictional force.

Final Answer: 147/10. Applying Forces & Equilibrium principles, $F = \mu N = 7/20 \times 42 = 147/10$ N Therefore, the verified correct option is 147/10.

3 [Foundational] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

A small particle of weight 24 N is held in equilibrium by two light inextensible strings. String A makes an angle of 30° with the upward vertical, and string B makes an angle of 60° with the upward vertical. Find the tensions T_A and T_B in the two strings.

Notice that the angle between string A and string B is $30^\circ + 60^\circ = 90^\circ$, so the strings are perpendicular. Resolve horizontally: $T_A \sin 30^\circ = T_B \sin 60^\circ \Rightarrow T_A(0.5) = T_B\left(\frac{\sqrt{3}}{2}\right) \Rightarrow T_A = \sqrt{3}T_B$ Resolve vertically: $T_A \cos 30^\circ + T_B \cos 60^\circ = 24 \Rightarrow T_A\left(\frac{\sqrt{3}}{2}\right) + T_B(0.5) = 24$ Substitute $T_A = \sqrt{3}T_B$.

4 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

A block of mass 8 kg rests in limiting equilibrium on a rough plane inclined at 30° to the horizontal. Taking $g = 9.8$ m/s^2: (a) calculate the normal reaction R, (b) find the coefficient of friction μ between the block and the plane.

(a) Resolve perpendicular to the inclined plane: $R = mg\cos 30^\circ = 8(9.8)\cos 30^\circ = 78.4 \times \frac{\sqrt{3}}{2} = 39.2\sqrt{3} \approx 67.9$ N (3 s. f.) (b) Resolve parallel to the plane. Since the block is in limiting equilibrium on the verge of sliding down, friction $F = \mu R$ acts up the plane: $F = mg\sin 30^\circ = 8(9.8)\sin 30^\circ = 78.4 \times 0.5 = 39.2$ N Setting $F = \mu R$.

5 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

[Forces & Equilibrium · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

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6 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

In a circle of radius 10 cm, chord AB subtends an angle of $\theta = \frac{2\pi}{3}$ radians at the centre O. Find: (a) the exact length of chord AB, (b) the exact area of the minor segment bounded by chord AB and minor arc AB, in terms of π and $\sqrt{3}$.

(a) Use the cosine rule in triangle OAB with $OA = OB = 10$ and $\angle AOB = \frac{2\pi}{3} = 120^\circ$: $AB^2 = 10^2 + 10^2 - 2(10)(10)\cos\left(\frac{2\pi}{3}\right) = 100 + 100 - 200\left(-\frac{1}{2}\right) = 200 + 100 = 300$ Thus $AB = \sqrt{300} = 10\sqrt{3}$ cm. (b) Area of sector OAB: $A_{\text{sector}} = \frac{1}{2}r^2\theta = \frac{1}{2}(10^2)\left(\frac{2\pi}{3}\right) = \frac{100\pi}{3}$ cm^2 Area of triangle OAB.

7 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

The circle C has equation $x^2 + y^2 - 6x + 4y - 12 = 0$. (a) Find the coordinates of the centre and the exact radius of C. (b) Find the equation of the tangent to C at the point $P(6, 2)$.

(a) Complete the squares for x and y: $(x^2 - 6x + 9) + (y^2 + 4y + 4) = 12 + 9 + 4 \Rightarrow (x - 3)^2 + (y + 2)^2 = 25$ The centre is $(3, -2)$ and the radius is $\sqrt{25} = 5$. (b) The gradient of the radius from centre $(3, -2)$ to $P(6, 2)$ is: $m_{\text{rad}} = \frac{2 - (-2)}{6 - 3} = \frac{4}{3}$ The tangent is perpendicular to the radius, so its gradient is $m_{\text{tan}} = -\frac{3}{4}$. Using point-slope form with $P(6, 2)$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Find the set of values of the constant k for which the equation $kx^2 + 4x + (k - 3) = 0$ has real roots.

For real roots, the discriminant $\Delta = b^2 - 4ac \geq 0$. Here $a = k$, $b = 4$, $c = k - 3$: $\Delta = 4^2 - 4(k)(k - 3) = 16 - 4k^2 + 12k \geq 0$ Divide the entire inequality by -4, reversing the inequality sign: $k^2 - 3k - 4 \leq 0$ Factor the quadratic: $(k - 4)(k + 1) \leq 0$. The critical values are $k = -1$ and $k = 4$. Since the parabola opens upward, the product is ≤ 0 between the roots: $-1 \leq k \leq 4$. Note that if $k = 0$, the equation becomes the linear equation $4x - 3 = 0$, which has one r.

9 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Find the equation of the normal to the curve $y = \frac{4}{\sqrt{x}} + 2x$ at the point where $x = 4$.

Rewrite in index form: $y = 4x^{-1/2} + 2x$. At $x = 4$: $y = 4(4^{-1/2}) + 2(4) = 4\left(\frac{1}{2}\right) + 8 = 2 + 8 = 10$. Point of contact is $(4, 10)$. Differentiate: $\frac{dy}{dx} = 4\left(-\frac{1}{2}x^{-3/2}\right) + 2 = -2x^{-3/2} + 2 = -\frac{2}{x^{3/2}} + 2$ At $x = 4$: $x^{3/2} = (\sqrt{4})^3 = 8$. Thus: $m_{\text{tan}} = -\frac{2}{8} + 2 = -\frac{1}{4} + 2 = \frac{7}{4}$ The normal has gradient $m_{\text{norm}} = -\frac{1}{7/4} = -\frac{4}{7}$. Using point $(4, 10)$.

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10 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)**

A bag contains 5 red and 4 green marbles. Three marbles are drawn at random without replacement. Find the probability that: (a) all three are red, (b) at least two are red.

Total marbles = 9. Number of ways to choose 3 marbles = $\binom{9}{3} = \frac{9\times8\times7}{3\times2\times1} = 84$. (a) All 3 are red: $\binom{5}{3} = 10$. Probability: $P(\text{all red}) = \frac{10}{84} = \frac{5}{42} \approx 0.119$ (b) 'At least two red' means exactly 2 red and 1 green, or exactly 3 red: Ways for 2 red, 1 green: $\binom{5}{2} \times \binom{4}{1} = 10 \times 4 = 40$. Ways for 3 red: $\binom{5}{3} = 10$. Total favorable outcomes: $40 + 10 = 50$. Probability.

11 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)**

A block of mass $4\ \text{kg}$ is pulled along a rough horizontal plane ($\mu = 0.25$) by a pulling force of magnitude $25\ \text{N}$ acting at 30° above the horizontal. Taking $g = 9.8\ \text{m/s}^2$: (a) show that the normal reaction $R = 26.7\ \text{N}$, (b) find the acceleration of the block.

(a) Resolve vertically (equilibrium in vertical direction): $R + 25\sin 30^\circ = mg \Rightarrow R + 25(0.5) = 4(9.8) \Rightarrow R + 12.5 = 39.2 \Rightarrow R = 26.7\ \text{N}$ (b) Limiting friction force opposes motion: $F = \mu R = 0.25 \times 26.7 = 6.675\ \text{N}$ Resolve horizontally using $F_{\text{net}} = ma$: $25\cos 30^\circ - F = ma \Rightarrow 25\left(\frac{\sqrt{3}}{2}\right) - 6.675 = 4a$.

12 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)**

The function f is defined by $f(x) = x^2 - 6x + 13$ for $x \geq 3$. (a) State the range of f. (b) Find an expression for $f^{-1}(x)$ and state the domain and range of f^{-1}.

(a) Complete the square: $f(x) = (x-3)^2 - 9 + 13 = (x-3)^2 + 4$. Since $x \geq 3$, $(x-3)^2 \geq 0$, so the minimum value is $f(3) = 4$. The range of f is $f(x) \geq 4$. (b) Set $y = (x-3)^2 + 4$. Rearrange for x: $y - 4 = (x-3)^2 \Rightarrow x - 3 = +\sqrt{y-4}$ (since $x \geq 3$) $x = 3 + \sqrt{y-4}$ Therefore, $f^{-1}(x) = 3 + \sqrt{x-4}$. Domain of f^{-1} is $x \geq 4$ (the range of f), and range of f^{-1} is $f^{-1}(x) \geq 3$ (the domain of f).

13 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

Functions g and h are defined by $g(x) = 2x - 1$ for $x \in \mathbb{R}$, and $h(x) = \frac{6}{x-2}$ for $x \neq 2$. (a) Find an expression for $hg(x)$ in simplest form. (b) Solve the equation $hg(x) = g(x)$.

(a) Substitute $g(x)$ into h: $hg(x) = h(2x-1) = \frac{6}{(2x-1)-2} = \frac{6}{2x-3} \quad \left(x \neq \frac{3}{2}\right)$ (b) Set $hg(x) = g(x)$: $\frac{6}{2x-3} = 2x - 1 \Rightarrow 6 = (2x-1)(2x-3)$ Expand the right-hand side: $6 = 4x^2 - 8x + 3 \Rightarrow 4x^2 - 8x - 3 = 0$ Factor: $(2x-3)(2x+1) = 0$. Since $x \neq \frac{3}{2}$ for $hg(x)$ to be defined, the value $x = \frac{3}{2}$ is excluded. Therefore, the unique solution is $2x + 1 = 0 \Rightarrow x = -\frac{1}{2}$.

14 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)**

Find the exact area of the region enclosed between the curve $y = 4x - x^2$ and the line $y = x$.

Find points of intersection: $4x - x^2 = x \Rightarrow 3x - x^2 = 0 \Rightarrow x(3-x) = 0$ The curves intersect at $x = 0$ and $x = 3$. Between 0 and 3, the curve $y = 4x - x^2$ lies above the line $y = x$. The enclosed area is: $A = \int_0^3 [(4x - x^2) - x]\,dx = \int_0^3 (3x - x^2)\,dx$ Integrate: $A = \left[\frac{3x^2}{2} - \frac{x^3}{3}\right]_0^3 = \left(\frac{3(9)}{2} - \frac{27}{3}\right) - 0 = \frac{27}{2} - 9 = \frac{27-18}{2} = \frac{9}{2} = 4.5$

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Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 **[Challenging]** • **Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)** • **Cross-Lesson: 9.01 Circular Measure**

A curve has equation $y = 2x^3 - 9x^2 + 12x + 1$. (a) Find the coordinates of both stationary points. (b) Determine the nature of each stationary point using the second derivative $\frac{d^2y}{dx^2}$.

(a) Find $\frac{dy}{dx}$: $\frac{dy}{dx} = 6x^2 - 18x + 12$ Set $\frac{dy}{dx} = 0$: $6(x^2 - 3x + 2) = 0 \Rightarrow 6(x - 1)(x - 2) = 0$, giving $x = 1$ and $x = 2$. When $x = 1$: $y = 2(1) - 9(1) + 12(1) + 1 = 6$. Point is $(1, 6)$. When $x = 2$: $y = 2(8) - 9(4) + 12(2) + 1 = 16 - 36 + 24 + 1 = 5$. Point is $(2, 5)$. (b) Second derivative: $\frac{d^2y}{dx^2} = 12x - 18$ At $x = 1$: $\frac{d^2y}{dx^2} = 12(1) - 18 = -60$, so $(2, 5)$ is a local minimum.

16 **[Challenging]** • **Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)** • **Cross-Lesson: 1.01 Indices, Surds & Quadratics**

The curve $y = x^2 - 6x + 5$ can be mapped to $y = x^2$ by a sequence of two geometric transformations. Express the equation in vertex form, state the vertex, and describe fully each transformation.

Complete the square: $y = (x - 3)^2 - 9 + 5 = (x - 3)^2 - 4$. The vertex of the parabola is $(3, -4)$. To map $y = (x - 3)^2 - 4$ back to $y = x^2$, we must translate the graph 3 units to the left and 4 units upwards. In vector notation, this is a single translation by $(-3; 4)$, or two sequential transformations: (1) Translation by vector $(-3; 0)$ (3 units left), and (2) Translation by vector $(0; 4)$ (4 units up).

17 **[Challenging]** • **Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)** • **Cross-Lesson: 10.01 Series & Binomial Theorem**

A car of mass 1200 kg ascends a hill of inclination α, where $\sin\alpha = \frac{1}{20}$, against a constant resistance to motion of 500 N. Taking $g = 9.8$ m/s^2: (a) find the driving force required to travel at a steady speed of 25 m/s, (b) calculate the power developed by the engine in kilowatts.

(a) Component of weight down the slope: $W_{\parallel} = mg\sin\alpha = 1200(9.8)(\frac{1}{20}) = 60(9.8) = 588$ N Since the car travels at steady speed (acceleration $a = 0$), the engine driving force balances all resistance: $F_{\text{drive}} = \text{Resistance} + mg\sin\alpha = 500 + 588 = 1088$ N (b) Power output: $P = F_{\text{drive}} \times v = 1088 \times 25 = 27\,200$ W $= 27.2$ kW

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18 **[Challenging]** • **Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)** • **Cross-Lesson: 11.01 Kinematics**

Events A and B are such that $P(A) = 0.6$, $P(B) = 0.5$, and $P(A \cup B) = 0.8$. Find $P(A \cap B)$, calculate $P(A|B)$, and determine whether A and B are independent.

1. Use the addition rule: $P(A \cup B) = P(A) + P(B) - P(A \cap B) \Rightarrow 0.8 = 0.6 + 0.5 - P(A \cap B) \Rightarrow P(A \cap B) = 1.1 - 0.8 = 0.3$. 2. Conditional probability: $P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{0.3}{0.5} = 0.6$ 3. Check for independence: $P(A \cap B) = 0.3$ and $P(A) \times P(B) = 0.6 \times 0.5 = 0.30$. Since $P(A \cap B) = P(A)P(B)$ (or equivalently $P(A|B) = P(A) = 0.6$), events A and B are independent.

19 **[Challenging]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)** • **Cross-Lesson: 13.01 Newton's Laws**

Find the equation of the perpendicular bisector of the line segment joining $A(-2, 5)$ and $B(4, -3)$. Give your answer in the form $ax + by + c = 0$, where a, b, c are integers.

1. Midpoint of AB: $M = \left(\frac{-2+4}{2}, \frac{5-3}{2}\right) = (1, 1)$. 2. Gradient of AB: $m_{AB} = \frac{-3-5}{4-(-2)} = \frac{-8}{6} = -\frac{4}{3}$. 3. Gradient of perpendicular bisector: $m_{\perp} = -\frac{1}{-4/3} = \frac{3}{4}$. 4. Equation of line passing through $M(1, 1)$: $y - 1 = \frac{3}{4}(x - 1) \Rightarrow 4(y - 1) = 3(x - 1) \Rightarrow 4y - 4 = 3x - 3$ Rearrange to standard integer form: $3x - 4y + 1 = 0$.

20 **[Challenging]** • **Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)** • **Cross-Lesson: 14.01 Energy, Work & Power**

The random variable $X \sim N(\mu, \sigma^2)$ satisfies $P(X < 65) = 0.8413$ and $P(X < 45) = 0.0228$. Find the values of μ and σ.

Standardising: $\Phi\left(\frac{65-\mu}{\sigma}\right) = 0.8413 \Rightarrow \frac{65-\mu}{\sigma} = 1.0$, so $65 - \mu = \sigma$. For the lower tail: $P(X < 45) = 0.0228 \Rightarrow \Phi\left(\frac{45-\mu}{\sigma}\right) = 0.0228 \Rightarrow \frac{45-\mu}{\sigma} = -2.0$, so $\mu - 45 = 2\sigma$. Adding the two equations: $65 - 45 = 3\sigma \Rightarrow 20 = 3\sigma \Rightarrow \sigma = 6.67$. Then $\mu = 65 - 6.67 = 58.33$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

A force of magnitude 50 N acts at an angle of 30° above the horizontal. Find its horizontal and vertical components.

a. Horizontal: 43.3 N**, Vertical:** 25.0 N ✓

b. Horizontal: 25.0 N, Vertical: 43.3 N

c. Horizontal: 35.4 N, Vertical: 35.4 N

d. Horizontal: 50.0 N, Vertical: 0.0 N

$F_x = 50\cos(30^\circ) = 50(\sqrt{3}/2) \approx 43.3$ N. $F_y = 50\sin(30^\circ) = 50(1/2) = 25.0$ N.

22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Two forces $F_1 = [8, -3]$ N and $F_2 = [-5, 7]$ N act on a particle. What is the magnitude of the resultant force?

a. 5 N ✓

b. 7 N

c. $\sqrt{41}$ N

d. 12 N

Resultant $\mathbf{R} = F_1 + F_2 = [8 - 5, -3 + 7] = [3, 4]$ N. Magnitude $|\mathbf{R}| = \sqrt{3^2 + 4^2} = 5$ N.

23 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

A particle is held in equilibrium by three coplanar forces: P acting East, Q acting North, and a force of 13 N acting at an angle θ South of West where $\tan\theta = 5/12$. Find the magnitude of P.

a. 12 N ✓

b. 5 N

c. 13 N

d. 8 N

For equilibrium, horizontal components balance: $P = 13\cos\theta$. Since $\tan\theta = 5/12$, $\cos\theta = 12/13$, so $P = 13(12/13) = 12$ N.

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24 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

A block of mass 6 kg rests on a rough horizontal table where the coefficient of friction is $\mu = 0.35$. Taking $g = 10$ m/s^2, what is the maximum frictional force that can act on the block?

a. 21 N ✓

b. 60 N

c. 18 N

d. 24 N

Normal reaction $R = mg = 6(10) = 60$ N. Maximum static friction $F_{\max} = \mu R = 0.35(60) = 21$ N.

25 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

A horizontal force of 15 N is applied to the 6 kg block in the previous question (where $F_{\max} = 21$ N). What is the actual frictional force acting on the block?

a. 15 N ✓

b. 21 N

c. 0 N

d. 6 N

Since the applied force (15 N) is less than the limiting friction (21 N), the block remains at rest and friction balances the applied force: $F = 15$ N.

26 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

A particle of mass 4 kg is placed on a rough plane inclined at 30° to the horizontal. If the particle is on the point of slipping down the plane, find the coefficient of friction μ.

a. $\mu = \tan(30^\circ) = \frac{\sqrt{3}}{3} \approx 0.577$ ✓

b. $\mu = \sin(30^\circ) = 0.5$

c. $\mu = \cos(30^\circ) \approx 0.866$

d. $\mu = 0.25$

At limiting equilibrium on an incline, $\mu = \tan\alpha$. For $\alpha = 30^\circ$, $\mu = \tan(30^\circ) = 1/\sqrt{3} = \sqrt{3}/3 \approx 0.577$.

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27 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

A crate of mass 10 kg is held in equilibrium on a smooth slope inclined at 30° to the horizontal by a force P acting parallel to the slope. Taking $g = 10$ m/s^2, find P.

a. 50 N ✓

b. 86.6 N

c. 100 N

d. 25 N

Resolving parallel to the slope: $P = mg\sin(30^\circ) = 10(10)(0.5) = 50$ N.

28 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For the crate in the previous question, what is the normal contact force exerted by the slope on the crate?

a. 86.6 N **(**$50\sqrt{3}$ N**)** ✓

b. 50 N

c. 100 N

d. 75 N

Resolving perpendicular to the slope: $R = mg\cos(30^\circ) = 100(\sqrt{3}/2) = 50\sqrt{3} \approx 86.6$ N.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A body of mass 8 kg rests in limiting equilibrium on an inclined plane. The normal reaction is 64 N. Taking $g = 10$ m/s^2, find the angle of inclination α of the plane.

a. $\alpha = \arccos(0.8) \approx 36.9^\circ$ ✓

b. $\alpha = 30^\circ$

c. $\alpha = 45^\circ$

d. $\alpha = 60^\circ$

$R = mg\cos\alpha \Rightarrow 64 = 8(10)\cos\alpha \Rightarrow 80\cos\alpha = 64 \Rightarrow \cos\alpha = 0.8 \Rightarrow \alpha \approx 36.9^\circ$.

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30 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A lamp of weight $W = 60$ N is suspended by two light strings making angles of 30° and 60° with the horizontal ceiling. Find the tension in the steeper string.

a. $30\sqrt{3}$ N ≈ 52.0 N ✓

b. 30 N

c. 60 N

d. 45 N

The angles with the vertical are 60° and 30°. The two strings are perpendicular ($30^\circ + 60^\circ = 90^\circ$). Resolving along the steeper string: $T_2 = W\cos(30^\circ) = 60(\sqrt{3}/2) = 30\sqrt{3}$ N ≈ 52.0 N.

31 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A light inextensible string passes over a smooth peg. A mass of 5 kg hangs on one side and a mass of 3 kg hangs on the other. What downward external force must be applied to the 3 kg mass to maintain equilibrium? ($g = 10$ m/s^2)

a. 20 N ✓

b. 50 N

c. 30 N

d. 10 N

Tension in string is $T = 5g = 50$ N. On the 3 kg side: $T = 3g + F \Rightarrow 50 = 30 + F \Rightarrow F = 20$ N.

32 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A ring of mass m is threaded on a rough horizontal wire with coefficient of friction $\mu = 0.4$. A light string attached to the ring is pulled with tension T at an angle of 45° above the wire. Find the value of T (in terms of mg) when the ring is in limiting equilibrium.

a. $T = \dfrac{0.4mg}{\cos(45^\circ) + 0.4\sin(45^\circ)} = \dfrac{0.4mg}{1.4\frac{\sqrt{2}}{2}} = \dfrac{4\sqrt{2}}{7}mg$ ✓

b. $\dfrac{\sqrt{2}}{2}mg$

c. $0.4mg$

d. mg

Resolving vertically: $R + T\sin(45^\circ) = mg \Rightarrow R = mg - T\sin(45^\circ)$. Horizontal friction: $F = \mu R = 0.4(mg - T\sin(45^\circ))$. For limiting equilibrium: $T\cos(45^\circ) = 0.4(mg - T\sin(45^\circ)) \Rightarrow T(\cos 45^\circ + 0.4\sin 45^\circ) = 0.4mg \Rightarrow T = \frac{0.4mg}{1.4(\sqrt{2}/2)} = \frac{4\sqrt{2}}{7}mg$.

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33 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

A particle of mass 5 kg rests on a smooth plane inclined at 45° to the horizontal. Taking $g = 9.8$ m/s^2, find the component of its weight acting down the slope, leaving your answer in exact (surd) form.

a. 49 N

b. $49/2 \times \frac{\sqrt{2}}{2}$ N

c. $51 \times \frac{\sqrt{2}}{2}$ N

d. $49 \times \frac{\sqrt{2}}{2}$ N ✓

$W = mg = 5 \times 9.8 = 49$ N. Component along slope $= W\sin 45^\circ = 49 \times \frac{\sqrt{2}}{2}$ N

34 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

A force of 10 N acts at 60° above the horizontal. Find its vertical component, leaving your answer in exact (surd) form.

a. $10 \times \frac{1}{2}$ N

b. $10 \times \frac{\sqrt{3}}{2}$ N ✓

c. 10 N

d. $5 \times \frac{\sqrt{3}}{2}$ N

Vertical component $= F\sin(60^\circ) = 10 \times \frac{\sqrt{3}}{2}$ N

35 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

A particle of mass 5 kg rests on a smooth plane inclined at 60° to the horizontal. Taking $g = 9.8$ m/s^2, find the normal reaction, leaving your answer in exact (surd) form.

a. $49 \times \frac{1}{2}$ N ✓

b. $49/2 \times \frac{1}{2}$ N

c. $49 \times \frac{\sqrt{3}}{2}$ N

d. 49 N

$W = mg = 5 \times 9.8 = 49$ N. $N = W\cos 60^\circ = 49 \times \frac{1}{2}$ N

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36 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

A force of 40 N acts at 60° above the horizontal. Find its horizontal component, leaving your answer in exact (surd) form.

a. 40 N

b. $20 \times \frac{1}{2}$ N

c. $40 \times \frac{\sqrt{3}}{2}$ N

d. $40 \times \frac{1}{2}$ N ✓

Horizontal component $= F\cos(60^\circ) = 40 \times \frac{1}{2}$ N

37 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Two perpendicular forces of magnitude 6 N and 8 N act on a particle. Find the magnitude of the resultant force.

a. 14

b. **10** ✓

c. 8

d. 12

$R = \sqrt{6^2 + 8^2} = \sqrt{100} = 10$ N

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 11.01 Kinematics

A sector has radius 5 cm and angle 0.8 rad. Find the perimeter of the sector.

a. **14** ✓

b. 9

c. 4

d. 15

$P = 2r + r\theta = 2(5) + (5)(0.8) = 14$ cm

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39 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 13.01 Newton's Laws

The scores on a standardized test are normally distributed with a mean of 100 and a standard deviation of 15. A new scoring algorithm is implemented, and the scores are transformed using the rule $Y = aX + b$, where X is the original score and Y is the new score. The new distribution of scores, Y, also follows a normal distribution with a mean of 120 and a standard deviation of 22.5. Calculate the value of a and the value of b.

a. a = 1.5, b = -30 ✓

b. a = -1.5, b = 270

c. a = 1.5, b = 30

d. a = 0.67, b = 53

Let $X \sim N(\mu_X, \sigma_X^2)$ and $Y \sim N(\mu_Y, \sigma_Y^2)$. We are given $\mu_X = 100$, $\sigma_X = 15$, $\mu_Y = 120$, and $\sigma_Y = 22.5$. The linear transformation is given by $Y = aX + b$. We know that for a linear transformation $Y = aX + b$, the mean of Y is $\mu_Y = a\mu_X + b$ and the standard deviation of Y is $\sigma_Y = |a|\sigma_X$. From the standard deviation relationship: $\sigma_Y = |a|\sigma_X$ $22.5 = |a| \times 15$ $|a| = \frac{22.5}{15} = 1.5$ Now consider the mean relationship.

40 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 14.01 Energy, Work & Power

The diagram shows the graph of the function $y = f(x)$. The function $f(x)$ is defined for all real numbers. Which of the following represents the graph of $y = f^{-1}(x)$?

a. Graph A

b. Graph B ✓

c. Graph C

d. Graph D

The graph of the inverse function $y = f^{-1}(x)$ is obtained by reflecting the graph of $y = f(x)$ across the line $y = x$. The provided graph of $f(x)$ is a parabola opening upwards with its vertex at $(2, -1)$. Graph A shows a function that is not a reflection of $f(x)$ across $y = x$. Graph B correctly shows the reflection of $y = f(x)$ across the line $y = x$. For example, the point $(0, 3)$ on $f(x)$ corresponds to the point $(3, 0)$ on $f^{-1}(x)$. The vertex $(2, -1)$ on $f(x)$ corresponds to the point.

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41 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 15.01 Representation of Data

A research team is studying the impact of different fertilizer types on crop yield. They conducted an experiment with two fertilizer types, A and B, and recorded the yield in kilograms per plot. The data is summarized in the two-way table below. The total number of plots is 200. | Fertilizer Type | Yield < 50 kg | Yield ≥ 50 kg | Total | |---|---|---|---| | A | 30 | 70 | 100 | | B | 50 | 50 | 100 | | Total | 80 | 120 | 200 | Given that a plot was selected at random and its yield was less than 50 kg, what is the probability that it received fertilizer type B? Provide your answer as a fraction in its simplest form.

a. 1/2

b. **5/8 ✓**

c. 3/8

d. 1/4

Let Y be the event that the yield is less than 50 kg, and B be the event that the fertilizer type is B. We are asked to find the conditional probability P(B|Y). From the table, we have: Number of plots with yield < 50 kg (event Y) = 80. Number of plots with yield < 50 kg AND fertilizer type B (event B ∩ Y) = 50. The formula for conditional probability is P(B|Y) = P(B ∩ Y) / P(Y). Alternatively, using the counts directly: P(B|Y) = (Number of plots with yield < 50 kg and fertilizer type B) / (Number of plot.

42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 16.01 Permutations & Combinations

Describe fully the single transformation that maps $y = x^2$ to $y = (x - 4)^2$.

a. Reflection in the x-axis

b. Translation in the y-direction only

c. A stretch parallel to the y-axis

d. **Translation by 4 unit(s) in the positive x-direction ✓**

Comparing the two equations term-by-term identifies the transformation as: Translation by 4 unit(s) in the positive x-direction.

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43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 17.01 Discrete Random Variables

A discrete random variable X has distribution $P(X = 1) = 2/5, P(X = 2) = 1/5, P(X = 3) = 2/5$. Given $Y = 5X - 3$, find $E(Y)$.

a. 6

b. **7 ✓**

c. 8

d. 13

Applying Discrete Random Variables principles, $E(X) = 2$. $E(Y) = aE(X) + b = 5(2) - 3 = 7$ Therefore, the verified correct option is 7.

44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 18.01 Binomial Distribution

Students are surveyed until the first one who owns a car is found; X is the number of students asked. Is X binomially distributed?

a. Yes — but only if $n > 30$

b. Yes — fixed, independent trials with constant success probability

c. No — the number of trials is not fixed

d. **No — trials are not independent / probability is not constant ✓**

The conditions for a binomial distribution (fixed independent trials with constant probability) are not satisfied here.

45 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 19.01 Normal Distribution

A machine's failure probability increases each hour it runs; X is the number of failures in a day. Is X binomially distributed?

a. **No — trials are not independent / probability is not constant ✓**

b. Yes — fixed, independent trials with constant success probability

c. No — the number of trials is not fixed

d. Yes — but only if $n > 30$

The conditions for a binomial distribution (fixed independent trials with constant probability) are not satisfied here.

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